

Tianyi ZHANG

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Education

MSc Precision Scientific Instrumentation	Expected 2026 – 2027
Nanyang Technological University	Singapore
- Focus on semiconductor instrumentation, precision electronics, sensing systems, advanced hardware integration, and artificial intelligence. - Recipient of an NTU postgraduate scholarship.	
BEng Mechanical Engineering, First Class Honours	Sept 2023 – Jun 2026
University College London	London, United Kingdom
Relevant coursework: control systems, computational modelling, algorithm development, embedded systems, and real-time systems.	

Skills

Programming: Python, Java, C++, MATLAB, Shell, Verilog
Semiconductor & Verification Tools: Synopsys VCS, Verdi, Linux, Perforce, RTL automation, regression testing
Software & Data Systems: FastAPI, Streamlit, SQL, Git, data processing pipelines, statistical forecasting
Engineering Tools: MATLAB Simulink, LabVIEW, ANSYS, Fusion 360

Work Experience

Assistant Project Engineer, MSIP Department, SRDC	Jul 2024 – Present
Advanced Micro Devices, Inc. (AMD)	Shanghai
- Contributed to memory read/write IP development for AMD accelerator projects. - Developed automation and AI-assisted verification scripts for large-scale SoC signal inspection and traffic analysis. - Built automated RTL error-injection frameworks to improve verification robustness and functional coverage evaluation. - Maintained regression and validation workflows using Linux, Shell scripting, Python, Perforce, Synopsys VCS, and Verdi. - Developed a real-time automated analytics platform to analyse code commits and measure AI-assisted contribution rates, improving visibility across engineering workflows. - Designed a context-aware skill-routing framework for AI-assisted Verilog debugging, replacing fixed single-skill iteration with dynamic scoring and selection based on error patterns, debugging stage, and design context.	

Projects

Evaluation-Driven Skill System for Software Engineering and EDA Automation LLM Agents / Evaluation Harness2026	
- Designed modular skill packages that encode domain workflows, tool-use instructions, and reusable execution patterns for software engineering and EDA automation tasks. - Built an evaluation framework to benchmark skill reliability across coding, debugging, RTL-analysis, and verification-assistance scenarios. - Created task datasets, scoring rubrics, and automated regression reports to compare baseline agent performance against skill-augmented workflows. - Integrated feedback from evaluation results into iterative prompt, workflow, and tool-invocation improvements for higher task-completion consistency.	
Energy-Aware Route Optimisation for Electric Scooters Graph Algorithms / OSM	2026
- Developed a simulation framework for electric scooter energy modelling using real-world map and elevation data. - Implemented graph-based routing algorithms with custom energy-consumption edge weights derived from terrain and traffic conditions. - Designed a hybrid optimisation strategy combining regenerative braking analysis with energy-aware route planning. - Built automated simulation pipelines for large-scale route evaluation and statistical comparison across urban scenarios.	
Path Planning for Trailer Obstacle Avoidance A* Algorithms / Kinematic Constraints	2023

- Developed a constrained path-planning framework for articulated trailer systems using an enhanced A* search algorithm.
- Integrated non-holonomic vehicle dynamics, steering continuity constraints, and collision-envelope detection into the planning heuristic.
- Generated dynamically feasible obstacle-avoidance trajectories while balancing computational efficiency and path optimality.

Full-Stack Immigration Data Management System | FastAPI / Streamlit / SQL

2025

- Developed a full-stack data management platform for the Singapore Immigration & Checkpoints Authority (ICA).
- Designed backend APIs and database structures using FastAPI for traveller record management and analytics.
- Built Streamlit dashboards for operational monitoring, data interaction, and visual analysis.
- Implemented short-term forecasting models to estimate future inbound traveller volume trends.
- Integrated automated data-processing pipelines to support decision-making workflows.

Leadership & Activities

UCL Racing Formula E Team

2023 – 2026

- Assisted in debugging motor signals and sensor feedback systems for the Formula E race car.

Vice President & Director of Program Control, Robotics Club

2020 – 2023

- Led the construction and debugging of robotic control modules and competed in the FRC China Regional Competition.

Honors & Awards

British Physics Olympiad Gold Award

2022

Additional Information

Languages: Mandarin Chinese (Native), English (IELTS 7.5)

Interests: Football, Table Tennis, Musicals, Travelling